ONCOLOGY · NEET-SS / FET / INI-SS

Giant cell and cystic lesions

1. Approach to a lytic (lucent) bone lesion

- **Four questions:** age of patient, location in the bone (epiphysis, metaphysis, diaphysis; central, eccentric, cortical), margin (zone of transition), matrix and periosteal reaction. Number of lesions.
- **Lodwick classification of margins (rate of growth):**

Grade	Pattern	Meaning
IA	Geographic, **sclerotic rim**	Slow growing, benign (NOF, UBC, fibrous dysplasia)
IB	Geographic, well-defined, **no sclerotic rim**	Faster (GCT, myeloma)
IC	Geographic, **ill-defined margin**	Aggressive
II	**Moth-eaten**	Aggressive (infection, lymphoma, metastasis)
III	**Permeative** (wide zone of transition)	Very aggressive: Ewing, lymphoma, osteomyelitis

- **Epiphyseal lucent lesions:** chondroblastoma (immature), **GCT** (mature), clear cell chondrosarcoma, infection, LCH, intraosseous ganglion, subchondral cyst (geode).
- **Multiple lesions in an adult:** metastases, myeloma, brown tumours, polyostotic fibrous dysplasia, enchondromatosis.
- **Fluid-fluid levels on MRI (not specific): ABC (primary or secondary), telangiectatic osteosarcoma**, GCT, chondroblastoma, UBC after fracture.

- **Giant-cell-rich lesions (histological differential):** GCT, **brown tumour of hyperparathyroidism**, ABC, chondroblastoma, NOF, giant cell reparative granuloma (jaw, hands and feet), giant-cell-rich osteosarcoma, pigmented villonodular synovitis extending into bone.
- **Rule:** a "benign-looking" cystic lesion with solid enhancing areas, thick nodular septa, cortical destruction or soft tissue mass needs a **biopsy before curettage** (telangiectatic OS can mimic ABC).

2. Giant cell tumour of bone (GCTB)

Epidemiology and pathology

- About **5% of primary bone tumours** and about 20% of benign bone tumours; **more common in Asian (including Indian and Chinese) populations**.
- **Age 20-40, after physeal closure** (about 80%); rare under 15 (if so, may be metaphyseal). **Slight female predominance**.
- **Sites: distal femur > proximal tibia > distal radius** (third commonest, and most aggressive with highest recurrence) > **sacrum** (commonest axial site; the vertebral body above the sacrum is uncommon) > proximal humerus. About 50-60% around the knee. Multicentric GCT is rare (under 1%; consider hyperparathyroidism and hand involvement).
- **Pathology:**
  - **Neoplastic cells = mononuclear stromal cells** (osteoblast-lineage), which express **RANKL**; they carry the **H3F3A (H3.3) G34W mutation in over 90%** (immunostain H3.3 G34W is diagnostic).
  - **Osteoclast-like giant cells** (expressing **RANK**) are reactive; they are recruited by RANKL and resorb bone. Their nuclei resemble the stromal cell nuclei.

    - **This RANKL-RANK pathway is the basis of denosumab therapy.**
- Classified by WHO as **intermediate, locally aggressive, rarely metastasising**.

Clinical

- Pain, swelling near a joint, effusion, reduced movement; **pathological fracture in about 10-30%**.

Imaging

- **Eccentric, lytic, epiphyseo-metaphyseal lesion extending up to the subchondral bone** (within about 1 cm of the articular surface).
- **Geographic, non-sclerotic margin (Lodwick IB)**, no matrix mineralisation, cortical expansion and thinning, "soap-bubble" appearance from ridges, cortical breach and soft tissue mass in aggressive cases. Periosteal reaction unusual.
- **MRI:** low-to-intermediate T2 signal in solid areas (haemosiderin and collagen), fluid-fluid levels from secondary ABC change (about 15%); defines soft tissue extension and joint involvement.
- **CT chest at diagnosis** (lung metastases).
- **Bloods: serum calcium, phosphate and PTH** to exclude **brown tumour of hyperparathyroidism**, especially with multiple lesions, unusual sites or giant cells in the jaw/hands.

Campanacci radiographic grading (corresponds to Enneking benign stages 1-3)

Grade	Radiographic features	Implication
I (latent)	Well-marginated, **thin sclerotic rim**, cortex intact	Rare; extended curettage

Grade	Radiographic features	Implication
II (active)	Well-defined margin **without sclerotic rim**, cortex thinned and expanded but intact	Commonest (about 70-80%); **extended curettage + adjuvant + cement**
III (aggressive)	**Indistinct margins, cortical breach, soft tissue mass**	Higher recurrence; curettage still possible if joint salvageable, otherwise **en bloc resection** and reconstruction

Management

Extended (intralesional) curettage: treatment of choice for most

1. **Large cortical window** (as big as the lesion) so every recess can be seen directly.
2. Thorough **curettage**, then **high-speed burr** of all walls.
3. **Local adjuvant** to extend the margin: **phenol**, **liquid nitrogen cryotherapy**, **argon beam coagulation**, hydrogen peroxide or absolute alcohol; pulsed lavage.
4. **Filling the cavity with PMMA bone cement**: exothermic polymerisation kills residual cells (about 2-3 mm), gives immediate structural support and allows early weight bearing, and makes **recurrence easy to detect** (expanding lucency at the cement-bone interface). Subchondral bone graft beneath the cement to protect articular cartilage is optional (debated). Internal fixation (plate) if the defect weakens the bone.
5. **Recurrence rates:** curettage alone historically **about 40-60%**; extended curettage with burr and adjuvant **about 10-20%**. Most recurrences appear **within 2-3 years**; distal radius, Campanacci III and soft tissue extension recur more often.
6. **Pathological fracture** is not an automatic indication for resection; if the fracture is not grossly displaced into the joint, curettage can be

done (after or with fixation).

En bloc (wide) resection

- **Indications:** extensive destruction with an unsalvageable joint, large soft tissue component (Campanacci III), grossly displaced intra-articular pathological fracture, multiply recurrent tumour, and **expendable bones** (proximal fibula, distal ulna, rib, iliac wing, coccyx).
- Lower recurrence (about 0-10%) but worse function.
- **Distal radius Campanacci III:** en bloc resection + reconstruction with **non-vascularised or vascularised proximal fibula autograft (osteoarticular, fibula head to the carpus)**, osteoarticular allograft, or **ulnar translocation with wrist arthrodesis** (radiocarpal fusion).
- Around the knee: megaprosthesis or osteoarticular allograft.

Spine and sacrum

- Sacral and spinal GCT are difficult: en bloc resection risks bowel, bladder and sexual dysfunction (sacral nerve roots).
- Options: **intralesional curettage + adjuvant with pre-operative embolisation**, **serial arterial embolisation**, **denosumab**, and **radiotherapy** (now reserved for unresectable disease because of the risk of **radiation-induced malignant transformation**, historically up to about 10%, lower with modern megavoltage techniques).

Medical therapy

- **Denosumab** (fully human monoclonal antibody against **RANKL**):
  - Dose: **120 mg subcutaneously every 4 weeks, with loading doses on days 8 and 15 of the first month**, with calcium and vitamin D supplements.
  - **FDA approved (2013)** for adults and skeletally mature adolescents with **unresectable GCT or where surgery would cause severe morbidity**, based on open-label phase 2 trials (Thomas et al., Lancet Oncology 2010; Chawla et al., 2013 and

2019) showing tumour response in most patients (loss of giant cells, new bone formation).
    - **Uses:** unresectable or sacral/spinal tumours, metastatic disease, and **neoadjuvant downstaging** to allow joint-preserving surgery (debated).
    - **Problems:**
    - **Higher local recurrence after curettage when used pre-operatively** (thick new bone and sclerotic rim hide residual tumour; stromal cells are not killed) - so for lesions already suitable for curettage, **go straight to surgery**.
    - **Recurrence or regrowth after stopping**, so long-term treatment may be needed (optimal duration unknown; dosing intervals are often extended).
    - **Osteonecrosis of the jaw** (about 1-5%, increasing with duration), **hypocalcaemia**, **atypical femoral fractures**, rebound hypercalcaemia in young patients after stopping, and **reports of malignant transformation** (sarcoma) during treatment.
- **Bisphosphonates (zoledronic acid)** also inhibit osteoclast giant cells; used as a cheaper adjuvant to reduce recurrence and for unresectable disease (evidence weaker).
- **Interferon alfa** and radiotherapy for refractory cases.

Metastases and malignancy

- **Lung metastases ("benign pulmonary implants") in about 2-5%**, usually 2-4 years after diagnosis and often after local recurrence (especially distal radius). Histologically identical to the primary; usually indolent. Treat by **metastasectomy**, observation or denosumab; good prognosis.
- **Malignancy in GCT (about 1-4%):**
    - **Primary malignant GCT:** high-grade sarcoma juxtaposed with conventional GCT at presentation.

  - **Secondary malignant GCT:** sarcoma at the site of previous GCT, usually **after radiotherapy** (or rarely after surgery or denosumab). Poor prognosis; treated like osteosarcoma.
- **Follow-up:** radiographs of the site and **chest imaging every 3-6 months for 2-3 years**, then annually to about 5-10 years.

> Exam pearl: GCT = skeletally mature, eccentric epiphyseo-metaphyseal lytic lesion reaching subchondral bone, non-sclerotic margin. Neoplastic cell = RANKL-expressing stromal cell (H3F3A G34W); giant cells are reactive. Distal femur > proximal tibia > distal radius (most aggressive); sacrum commonest axial site. Extended curettage + burr + adjuvant + PMMA (recurrence about 10-20%). Denosumab for unresectable disease - not routinely before curettage (more recurrence). Lung implants 2-5%. Always check calcium/PTH to exclude brown tumour.

3. Unicameral (simple) bone cyst (UBC)

Features

- Fluid-filled cavity lined by a thin fibrous membrane; **not a true neoplasm** (although **NFATC2 fusions** have recently been described). Cause thought to be **venous obstruction** with fluid rich in prostaglandins, interleukin-1 and enzymes that resorb bone.
- **Age 5-15 years, male to female about 2:1**.
- **Sites: proximal humerus (about 50-60%)**, **proximal femur (about 25-30%)**; in adults, **calcaneus** (neutral triangle under the angle of Gissane) and ilium.
- **Pathological fracture is the commonest presentation (about two thirds to three quarters)**; otherwise incidental.

Radiology

- **Central (not eccentric) metaphyseal lytic lesion**, **symmetrically expanded but not wider than the physis**, thin sclerotic margin, thinned cortex, no periosteal reaction without fracture, no matrix.
- **"Fallen fragment" sign** (a cortical fragment sinks to the dependent part of the fluid-filled cyst) is **pathognomonic** of UBC (versus solid lesions). The "rising bubble" sign is the gas equivalent.
- **Active vs latent:**

Phase	Features	Prognosis
Active	**Within about 1 cm of (abutting) the physis**, usually **under 10-12 years**	High recurrence after treatment; risk of physeal growth arrest
Latent (inactive)	**Migrated away from the physis** into the diaphysis by normal growth (over 1 cm), older child	Lower recurrence; tends to heal

- **Risk of fracture:** cyst index (cyst area / diaphyseal diameter squared) and thin cortex; fracture is likely when the cyst occupies most of the bone width.
- **Calcaneal UBC versus intraosseous lipoma:** lipoma has fat density and often a **central calcification ("cockade" / target)**; UBC has fluid content.
- **Differentials:** ABC (eccentric, expands wider than the physis, fluid-fluid levels, multiple septa), fibrous dysplasia (ground glass), LCH, enchondroma, NOF.

Management

- **Fracture through a proximal humeral cyst:** treat the fracture non-operatively (sling); the fracture heals in about 6 weeks, but **the cyst heals after fracture in only about 10-15%**. Then decide on treatment of the cyst.
- **Observation** in older children with latent lesions and low fracture risk.

- **Minimally invasive options:**
  - **Aspiration + intralesional methylprednisolone acetate injection** (Scaglietti 1979); often **multiple injections** needed. Confirm with straw-coloured fluid and contrast cystogram (contrast stays in a single cavity).
  - Injection of **autologous bone marrow** or **demineralised bone matrix**, or calcium sulphate pellets.
  - **Decompression** (trephination / curettage with disruption of the cyst wall and canal communication) and **flexible intramedullary nails** (elastic stable intramedullary nails act as a continuous drain and give stability; allow early return to activity), or cannulated screws.
- **Evidence: Wright et al. (JBJS Am 2008), multicentre RCT: methylprednisolone injection gave a higher healing rate (about 42%) than bone marrow injection (about 23%)** at 2 years. Overall, no technique gives complete healing in every case; recurrence after any method is common (about 20-40%).
- **Open curettage and bone grafting** with or without adjuvant: for failed injections or large lesions; risk of **physeal injury** in active cysts.
- **Proximal femoral UBC:** high risk of fracture with **varus malunion, AVN and growth arrest** - so treat **surgically**: curettage + graft and **internal fixation** (flexible nails, pediatric hip screw or plate, avoiding the physis), especially for large cysts or after fracture.
- **Assessment of healing** (Neer/Cole type classifications): healed, healed with residual radiolucency, persistent, recurrent.

> Exam pearl: UBC = 5-15 years, proximal humerus, central metaphyseal, not wider than the physis, fallen fragment sign pathognomonic. Active = abutting the physis (under 1 cm), recurs more. Fracture heals but the cyst rarely heals. Steroid injection beat bone marrow injection in the Wright 2008 RCT. Proximal femur cysts need surgical fixation.

4. Aneurysmal bone cyst (ABC)

Features

- **Blood-filled, multiloculated expansile lesion**; WHO intermediate, locally aggressive.
- **Age under 20 in about 80%** (peak 10-20), roughly equal sex distribution.
- **Sites:** metaphysis of long bones (**femur, tibia, humerus**), **spine (posterior elements, can extend into the body; about 15-20%)**, pelvis, sacrum.
- **Primary ABC (about 70%)**: true neoplasm with **USP6 gene rearrangement** (17p13; commonest partner **CDH11**). Also seen in **nodular fasciitis and myositis ossificans** (USP6 family).
- **Secondary ABC (about 30%)**: ABC-like change within another lesion, without USP6 rearrangement - **GCT** (commonest), **chondroblastoma**, **osteoblastoma**, fibrous dysplasia, NOF, UBC, and **telangiectatic osteosarcoma**. Treatment is that of the underlying lesion.
- **Solid variant (giant cell reparative granuloma-like)** is part of the same spectrum.

Clinical and imaging

- Pain and swelling of short duration, pathological fracture; spine: back pain, scoliosis, **radiculopathy or cord compression**.

- **Radiograph: eccentric, expansile, "blow-out" lytic lesion** with a thin **eggshell** cortical rim, internal trabeculation ("soap bubble"), may expand **wider than the physis**; Codman triangle may be seen at the margins.
- **MRI: multiple fluid-fluid levels** (blood products of differing ages), **thin, smooth enhancing septa**.
- **Telangiectatic osteosarcoma must be excluded:** thick nodular enhancing septa, solid component, cortical destruction, soft tissue mass and aggressive margins favour TOS. **Biopsy of the septa and solid areas** before definitive treatment.
- **Histology:** blood-filled cavernous spaces separated by **fibrous septa containing fibroblasts, osteoclast-like giant cells and "blue bone"** (reactive woven bone); no endothelial lining.

Management

Option	Notes
Extended curettage + adjuvant (burr, phenol, cryotherapy, argon) + bone graft or cement	Standard; **recurrence about 10-20%**, higher in young children and lesions near the physis; most recurrences within 2 years
Percutaneous sclerotherapy (**polidocanol**, doxycycline, alcohol)	Minimally invasive; multiple injections; an RCT (**Varshney et al. 2010**) found polidocanol gave healing comparable to curettage with fewer complications; increasingly used first line, especially in difficult sites
Selective arterial embolisation	Pelvis, sacrum and spine; as primary treatment or to reduce intra-operative bleeding before surgery
Denosumab	Unresectable spinal or sacral lesions; regrowth after stopping and **rebound hypercalcaemia in children** are concerns

Option	Notes
En bloc resection	Expendable bones (fibula, rib) or recurrent aggressive lesions
Spine	Pre-operative embolisation, curettage or excision, decompression and **instrumented stabilisation** for instability or deficit
Radiotherapy	**Avoided** (risk of radiation-induced sarcoma, growth disturbance); last resort

> Exam pearl: Primary ABC = USP6 (CDH11-USP6) rearrangement; secondary ABC commonest with GCT. Eccentric, expansile, eggshell rim, fluid-fluid levels. Must exclude telangiectatic osteosarcoma (thick nodular septa) by biopsy. Curettage + adjuvant recurs 10-20%; polidocanol sclerotherapy is an accepted alternative.

5. UBC versus ABC versus GCT

Feature	UBC	ABC	GCT
Age	5-15	10-20	20-40 (mature)
Position	Central metaphysis	Eccentric metaphysis	Eccentric epiphyseo-metaphyseal to subchondral bone
Expansion	Not wider than physis	Blow-out, wider than physis	Expanded, thin cortex
Margin	Sclerotic, well-defined	Thin eggshell rim	Non-sclerotic, geographic
MRI	Single fluid cavity	Multiple fluid-fluid levels	Solid, low T2 areas
Genetics	NFATC2 fusions (recent)	USP6 rearrangement	H3F3A G34W

Feature	UBC	ABC	GCT
Key sign	Fallen fragment	Fluid-fluid levels	Reaches subchondral bone

6. Fibrous and other cystic-looking lesions (common distractors)

Non-ossifying fibroma (NOF) and fibrous cortical defect

- **Commonest bone lesion in children** (fibrous cortical defects in about 30% of children); age 5-20; **"don't touch" lesion**.
- **Distal femur and distal or proximal tibia metaphysis**; **eccentric, cortically based, multiloculated lucency with a scalloped sclerotic margin**; long axis parallel to the bone; migrates away from the physis and **heals spontaneously by sclerosis** in adolescence.
- Histology: storiform fibroblasts, giant cells, haemosiderin and foam cells; **KRAS / FGFR1 (MAPK pathway)** mutations.
- **Fracture risk (Arata):** lesion involving **over 50% of the transverse diameter on AP and lateral views and over 33 mm long**.
- **Treatment:** observation; pathological fractures usually heal in a cast; curettage and grafting for large symptomatic lesions or after fracture with persistent risk.
- **Jaffe-Campanacci syndrome:** multiple NOFs + café-au-lait spots (+ learning difficulty, hypogonadism, ocular and cardiac anomalies); now considered part of **neurofibromatosis type 1** spectrum.

Fibrous dysplasia (FD)

- **Activating GNAS mutation (Gs alpha, codon R201)**, post-zygotic somatic mosaic - **not inherited**; increased cAMP leads to immature fibro-osseous tissue.
- **Monostotic (about 70-80%): ribs (commonest), proximal femur, craniofacial bones**, tibia. **Polyostotic:** often unilateral, femur, tibia, pelvis.

- **McCune-Albright syndrome:** polyostotic FD + **café-au-lait spots with irregular "coast of Maine" borders** (NF1 spots have smooth "coast of California" borders) + **endocrinopathy** (**precocious puberty** commonest, especially girls; hyperthyroidism, growth hormone excess, Cushing syndrome) and **FGF23-mediated renal phosphate wasting (hypophosphataemic rickets/osteomalacia)**.
- **Mazabraud syndrome:** FD + **intramuscular myxomas**.
- **Radiograph: ground-glass** matrix, well-defined lesion with a **thick sclerotic "rind"**, expansion, endosteal scalloping; long lesion in the diaphysis/metaphysis; **shepherd's crook deformity** (coxa vara of the proximal femur); bowing; leg-length discrepancy.
- **Histology:** irregular curvilinear woven bone trabeculae (**"Chinese letters" / "alphabet soup"**) in fibrous stroma, **without osteoblastic rimming**.
- **Management:**
  - Observation for asymptomatic lesions; **bisphosphonates** (IV pamidronate/zoledronate) reduce pain (do not heal the lesion).
  - **Treat hypophosphataemia** and endocrine disease before surgery.
  - **Autogenous cancellous graft is resorbed and replaced by dysplastic bone**; use **cortical allograft struts** (resorb slowly) if grafting.
  - **Proximal femur / shepherd's crook: valgus osteotomy with intramedullary fixation (cephalomedullary nail)** - load-sharing intramedullary devices are preferred over plates in children and for long-term protection.
  - Malignant transformation (osteosarcoma, fibrosarcoma) **under 1%**, higher in McCune-Albright and after **radiotherapy (avoid)**.

Osteofibrous dysplasia (OFD, Campanacci lesion) and adamantinoma

Feature	Osteofibrous dysplasia	Adamantinoma
Age	**Under 10 years**	20-40 years
Site	**Anterior cortex of the tibial diaphysis** (also fibula); anterior bowing	**Anterior tibial diaphysis** (about 85-90% in the tibia)
Radiograph	Intracortical multiloculated lucencies with sclerosis	Multiloculated "soap-bubble" lytic lesion, may breach cortex, satellite lesions
Histology	Woven bone **with osteoblastic rimming** (unlike FD), scattered keratin-positive cells	**Epithelial (keratin-positive) nests** in fibro-osseous stroma
Nature	Benign; may regress after puberty	**Low-grade malignant**; late lung and node metastases (about 15-30%)
Treatment	**Observation**; bracing for bowing; delay surgery until after about age 15 (early resection recurs)	**Wide en bloc resection** with intercalary reconstruction (allograft, vascularised fibula, bone transport) or amputation; chemo- and radio-resistant

- **OFD-like (differentiated) adamantinoma** is an intermediate entity; OFD and adamantinoma are thought to form a spectrum, so biopsy aggressive or enlarging OFD-like lesions.

Langerhans cell histiocytosis (eosinophilic granuloma)

- Clonal proliferation of Langerhans-type dendritic cells; **BRAF V600E mutation in about 50%**.
- **Age 5-15 (most under 30)**; **"great mimicker"** (of Ewing sarcoma and osteomyelitis).
- **Skull is the commonest site** (**punched-out lesion with bevelled edges**, "hole within a hole", button sequestrum), **vertebra plana**

(thoracic spine, collapse with preserved discs; Calvé disease), ribs, pelvis, long bone diaphysis (may show periosteal reaction).

- **Histology:** Langerhans cells with **grooved "coffee-bean" nuclei**, eosinophils; **CD1a, CD207 (langerin) and S100 positive**; **Birbeck granules ("tennis racquet")** on electron microscopy.
- **Clinical forms:** unifocal bone (eosinophilic granuloma, commonest, best prognosis); **Hand-Schüller-Christian** (multifocal: **skull lesions, exophthalmos, diabetes insipidus**); **Letterer-Siwe** (disseminated multisystem disease under 2-3 years, poor prognosis).
- **Treatment:** skeletal survey/PET to assess spread; unifocal lesions often **heal spontaneously** - biopsy, then observation, **intralesional methylprednisolone**, or curettage; **vertebra plana**: observation ± brace (partial height restoration in children), surgery only for neurological deficit or instability; low-dose radiotherapy rarely; **multisystem disease: chemotherapy (vinblastine + prednisolone)**, BRAF inhibitors in refractory cases.

Other lucent lesions

- **Brown tumour (osteitis fibrosa cystica):** hyperparathyroidism; multiple lytic lesions, subperiosteal resorption of the **radial side of middle phalanges**, salt-and-pepper skull; giant cells on histology mimic GCT; **resolve after parathyroidectomy**.
- **Intraosseous ganglion:** subchondral, adults, medial malleolus, around the knee and carpal bones; sclerotic rim; curettage if symptomatic.
- **Epidermoid inclusion cyst: distal phalanx** or skull after penetrating trauma; well-defined lytic lesion; curettage.
- **Intraosseous lipoma:** calcaneus (central calcification), proximal femur; observation.
- **Mnemonic for lucent lesions (FEGNOMASHIC):** Fibrous dysplasia, Enchondroma / Eosinophilic granuloma, GCT, NOF, Osteoblastoma, Metastasis / Myeloma, ABC, Simple bone cyst, Hyperparathyroidism

(brown tumour), Infection, Chondroblastoma / Chondromyxoid fibroma. Lesions not likely over 30 years: EG, NOF, ABC, SBC, chondroblastoma; GCT needs a closed physis.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.